

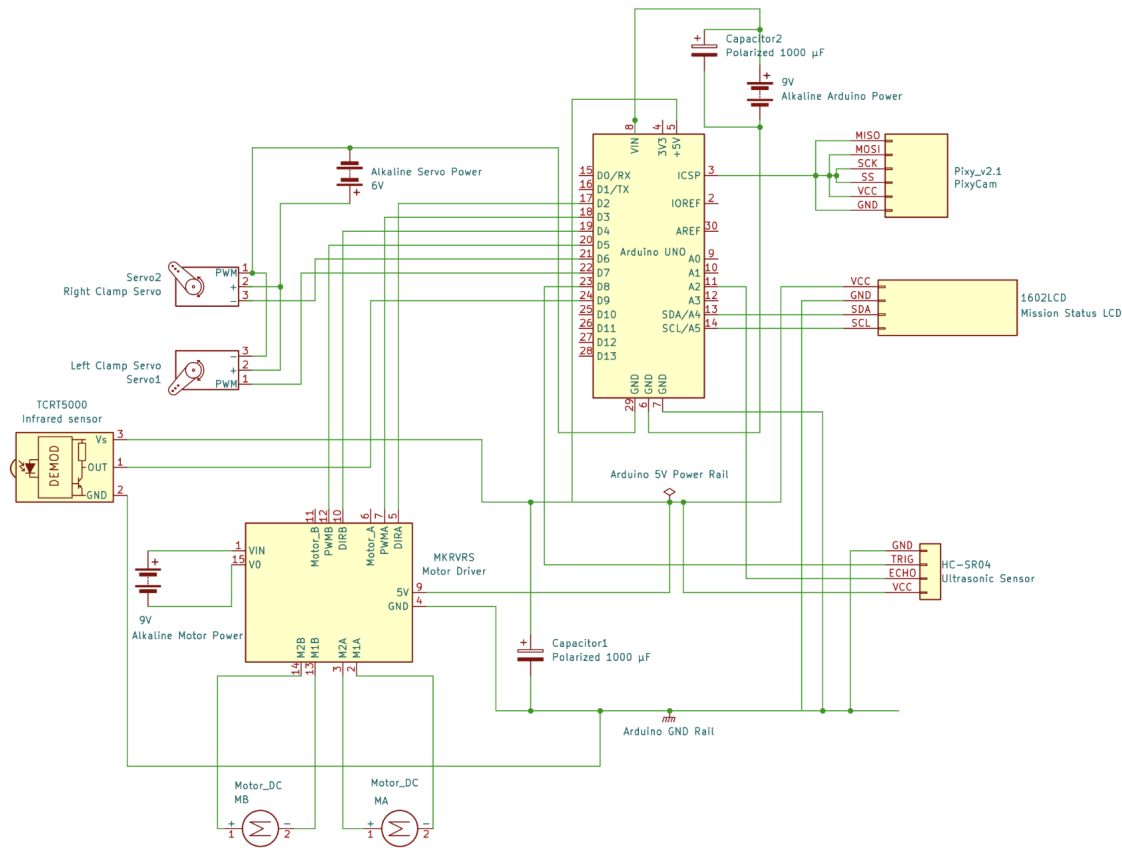


Figure 7: Circuit Diagram of the robot, without perfboard and SPDT switch.

1.3.3 Software Design

The software is implemented on the Arduino Uno using a modular structure with clear task separation.³ The `setup()` routine initializes sensors and actuators, logs the base signature at startup, and establishes a known mechanical state. The `loop()` routine then executes repeated colour cycles in the required order. Each cycle calls dedicated functions for search, centring, chase, clamp actuation, return-to-base, and deposit, so motion control and decision logic remain consistent across all three target colours. Operational reliability is ensured through rule-based checks and feedback at each stage. PixyCam 2.1 block position and size are used for steering and approach control, the IR sensor confirms capture before transition to return mode, and ultrasonic readings support obstacle handling during normal navigation.^{5,7,10} Timeout limits, confirmation checks, and orientation state tracking are also applied to reduce false transitions and prevent repeated turning errors, which helps the robot recover from temporary loss of visual targets and continue the mission sequence. See Section 1.4.5, Subsystem Integration for more details.

1.4 Subsystems and Integration

1.4.1 Control Subsystem

The Control Subsystem manages the overall operation of the robot.³ It consists of an Arduino Uno microcontroller and an LCD display, which provides simple user feedback.

The subsystem operates by the Arduino receiving sensor data from the Vision and Detection Subsystem. Specifically, the PixyCam 2.1 detects coloured balls and the robot's base, the HC-SR04 ultrasonic sensor